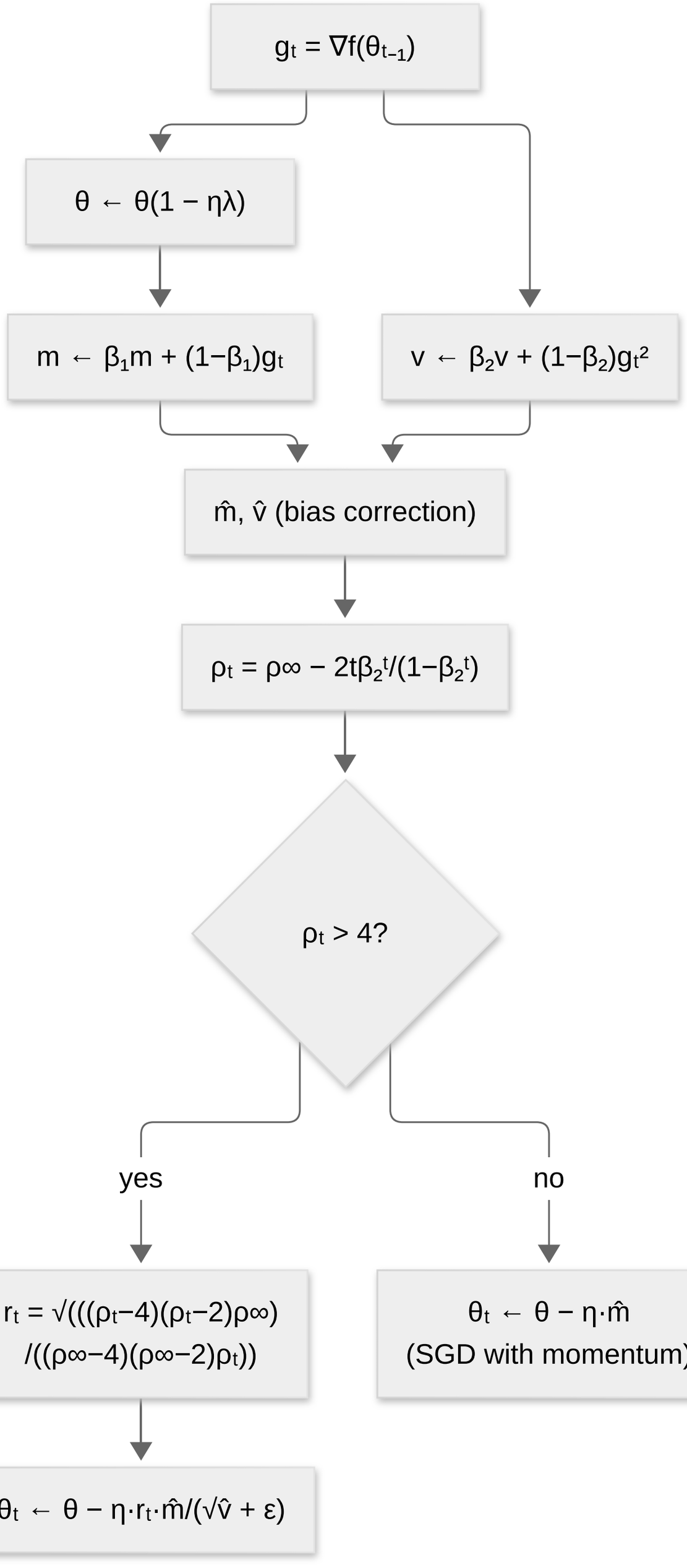

$g_t = \nabla f(\theta_{t-1})$
$\theta \leftarrow \theta(1 - \eta\lambda)$
$m \leftarrow \beta_1 m + (1-\beta_1) g_t$
$v \leftarrow \beta_2 v + (1-\beta_2) g_t^2$
$\hat{m}$, $\hat{v}$ (bias correction)
$\rho_t = \rho_\infty - 2t\beta_2^t/(1-\beta_2^t)$
$\rho_t > 4$?
yes
no
$r_t = \sqrt{((\rho_t-4)(\rho_t-2)\rho_\infty)/((\rho_\infty-4)(\rho_\infty-2)\rho_t)}$
$\theta_t \leftarrow \theta - \eta \cdot r_t \cdot \hat{m}/(\sqrt{\hat{v}} + \varepsilon)$
$\theta_t \leftarrow \theta - \eta \cdot \hat{m}$
(SGD with momentum)